Understanding Data Before Analysis:

Software/ML Domain — Task 7

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I. Introduction

Before any dataset can be cleaned, visualised, or modelled, it has to be understood: what each row represents, what each column measures, and what unit and context give that measurement meaning. This report addresses the core concepts of observations, variables, categorical and numerical data, raw and processed data, measurement units, and basic summary statistics, using examples from Biochemistry, Electronics, and Mechanical domain data.

II. Observation vs. Variable

An observation is an individual unit being reviewed or measured, whereas a variable is a property or attribute that is a part of the observation. In the context of data analysis, an observation is a singular data point — one specific instance — whereas a variable is a specific property recorded for that data point.

Example: an observation is one specific student's record, while a variable is his enrollment number — one property of that student's record.

III. Categorical vs. Numerical Data

Categorical data represents values that fall into a fixed set of groups, describing "which kind" or "which category" something belongs to, rather than a measurable quantity. Numerical data, on the other hand, consists of values that represent an actual measurable quantity and can be used in arithmetic operations such as averaging or comparison.

Example: Categorical data — submission status (e.g., "Submitted" / "Not Submitted"). Numerical data — marks (e.g., 85, 92, 76).

IV. Domain Examples of Categorical and Numerical Data

One categorical and one numerical example from each of the three assigned domains are given below, along with the unit each numerical example is expressed in.

Table 1. Categorical and numerical data examples across three domains

Domain	**Categorical Example**	**Numerical Example**	**Why Unit/Context Matters**
Biochemistry	Type of microorganism used in an experiment (e.g., bacteria vs. yeast)	Enzyme concentration (mg/mL)	An enzyme concentration of "5" only has meaning as 5 mg/mL; without the unit it could be mistaken for a different quantity such as 5 mol/L.
Electronics	Type of circuit (e.g., analog vs. digital)	Frequency (Hz)	A frequency of "50" is meaningless without knowing it is 50 Hz; the same numeral could

			otherwise be confused with 50 V or 50 Ω in the same circuit dataset.
Mechanical	Type of joint used in a structure (e.g., welded vs. bolted)	Torque (Nm)	A torque of "120" only makes sense as 120 Nm; treating it as a bare number could confuse it with force (N) or displacement (mm), which are dimensionally different quantities.

V. Domain Thinking: Why Unit and Context Cannot Be Dropped

A dataset is not simply a table of numbers. A value only becomes meaningful once it is connected to its variable name, its measurement unit, the condition under which it was recorded, and the domain it belongs to. The table below shows the minimum set of domain examples illustrating this principle.

Table 2. Minimum required domain examples

Domain	**Categorical Example**	**Numerical Example**	**Why Unit/Context Matters**
Biochemistry	Condition: control or heated	pH, concentration, absorbance	pH 7 and 7 volts are not comparable values.
Electronics	Load type: low load or high load	voltage, current, resistance	5 V and 5 mA represent different physical quantities.
Mechanical	Material type: A or B	force, strain, displacement	120 N means force, not score or percentage.

VI. Raw Data vs. Processed Data

Raw data refers to a dataset in its original form, before any kind of analysis or cleaning has been done — it has the data exactly as it was collected. Processed data, on the other hand, refers to a dataset that has been analysed and cleaned to handle outliers and missing values, resulting in more uniformity and consistency.

VII. Measurement Units and Their Importance

A measurement unit is a standard scale assigned to a numerical value based on the variable it represents. It is extremely crucial in data analysis because it clearly differentiates values based on what they actually represent, allowing data to be processed correctly and preventing false equivalence — incorrect comparisons between values that happen to share the same number but represent entirely different quantities.

VIII. Danger of Comparing Values Across Units

It is extremely dangerous to compare values from different units directly because, regardless of their numerical value, they hold entirely different meanings within the dataset. Comparing values across different units can create a false sense of relevance or correlation in the data, which could lead to flawed conclusions or errors when the data is later used to train a model.

IX. Summary Statistics: Mean, Median, Minimum, Maximum, and Range

Mean refers to the overall average of a variable, calculated by summing all its values and dividing by the number of values. Median refers to the middle value of a variable when all its values are arranged in order. Minimum and maximum refer to the lowest and highest numerical values of a variable, respectively. Range refers to the difference between the maximum and minimum values of a variable, giving a measure of how spread out the data is.

X. When the Mean Can Be Misleading

The mean can be misleading in the presence of outliers — extremely high or extremely low values that are far from the rest of the data. Since the mean accounts for every value in the calculation, even one or two extreme values can pull it significantly away from what most of the data actually looks like, giving a distorted picture of the "typical" value.

XI. Checks Required Before Analysing a Dataset

Before doing any analysis on a dataset, the following should be inspected:

- Missing data — checking for null or absent values that could affect the analysis.
- Type mismatch — verifying that each variable's data type (numerical, categorical, etc.) is consistent and correctly assigned.
- Inconsistent representation of a single variable — checking whether the same variable is recorded in multiple different formats or units across observations.

XII. Questions to Ask Before Analysing Domain Data From Another Team

When domain data is received from another team, the following questions should be asked:

- Has the data already been cleaned/pre-processed, or is this raw data?
- What data cleaning methods were followed?
- Was data normalization carried out?
- What are the standard units used for each variable, and are they clearly mentioned in the variable headers?

XIII. Conclusion

A dataset is not simply a table of numbers; each value is only meaningful once it is tied to its variable name, its measurement unit, the condition under which it was recorded, and the domain it comes from. Establishing this understanding — distinguishing observations from variables, categorical from numerical data, and raw from processed data, while recognising the limits of summary statistics such as the mean — is the necessary first step before any dataset from Biochemistry, Electronics, or Mechanical domains can be responsibly analysed.

References

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